Industrial Vision

Lecture 15 - Algebraic Epipolar Geometry

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Where we are

We obtained a **procedure** for constructing the epipolar line in the right view corresponding to a point $\mathbf{x}$ in the left view. A ray is back-projected, **two** points are taken on it, both are projected using P_R, and they are joined by a **cross product**. The match then satisfies $\ell'^{\top}\tilde{\mathbf{x}}' = 0$.

Today we carry out that procedure symbolically and find that it reduces to a single 3×3 matrix:

$$\ell' = F\,\tilde{\mathbf{x}}, \qquad \boxed{\tilde{\mathbf{x}}'^{\top} F\,\tilde{\mathbf{x}} = 0}$$

The matrix F, called the **fundamental matrix**, encodes the entire epipolar geometry of the camera pair. Its calibrated counterpart is E, the **essential matrix**. Two preliminary tools are required: the **skew-symmetric** form of the cross product, and the **back-projected ray**.

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The first tool: the cross product written as a matrix

Any cross product $\mathbf{v} \times \mathbf{y}$ may be written as the product of the skew-symmetric matrix $[\mathbf{v}]_\times$ with a vector:

$$\mathbf{v} \times \mathbf{y} = [\mathbf{v}]_\times \, \mathbf{y}, \qquad [\mathbf{v}]_\times = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}$$

Properties. The matrix $[\mathbf{v}]_\times$ is skew-symmetric and has **rank 2**, and $\mathbf{v}$ is its **null vector**, since $[\mathbf{v}]_\times \mathbf{v} = \mathbf{v} \times \mathbf{v} = \mathbf{0}$.

An identity that will be required, concerning a matrix passed through a cross product:

$$(M\mathbf{a}) \times (M\mathbf{b}) = \det(M)\, M^{-\top} (\mathbf{a} \times \mathbf{b}).$$

The identity is often written as $M\mathbf{a} \times M\mathbf{b} = M^{-\top}(\mathbf{a} \times \mathbf{b})$, with the factor $\det M$ omitted. As F is defined **only up to scale** and the scalar $\det M$ is absorbed into the overall factor of F.

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The second tool: the back-projected ray

Consider a point in the left image with actual pixel coordinates $\tilde{\mathbf{x}} = (x, y, 1)^\top$. Removing the effect of the intrinsic parameters, the quantity $K_L^{-1}\tilde{\mathbf{x}}$ is a **direction**. Scaling it by a depth z and expressing the result in homogeneous three-dimensional coordinates gives

$$X(z) = \begin{bmatrix} z\,K_L^{-1}\tilde{\mathbf{x}} \\ 1 \end{bmatrix} \quad (4 \times 1) \qquad \text{the ray from } C \text{ through } \tilde{\mathbf{x}}.$$

Verification that this is the ray. Applying $P_L = K_L[\,I \mid \mathbf{0}\,]$ to $X(z)$ gives

$$P_L\,X(z) = K_L\,[\,I \mid \mathbf{0}\,] \begin{bmatrix} zK_L^{-1}\tilde{\mathbf{x}} \\ 1 \end{bmatrix} = K_L\,(zK_L^{-1}\tilde{\mathbf{x}}) = z\,\tilde{\mathbf{x}} \xrightarrow{\div z} \tilde{\mathbf{x}}.$$

Every value of z maps back to the same image point (x, y), which is precisely what it means for the points to lie on the back-projected ray.

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Two convenient points on the ray

Any two points on the ray will serve, and we choose the two that simplify the algebra most, namely those at $z = 0$ and at $z = \infty$.

At $z = 0$ the point is the camera center C:

$$X(0) = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} = (0, 0, 0, 1)^\top.$$

At $z = \infty$ the point is the point at infinity along the ray, for which the last coordinate tends to zero:

$$X(\infty) = \begin{bmatrix} K_L^{-1}\tilde{\mathbf{x}} \\ 0 \end{bmatrix}.$$

Both points are projected into the right image using $P_R = K_R[\, R \mid \mathbf{t} \,]$:

$$P_R\, X(0) = K_R[\, R \mid \mathbf{t} \,] \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix} = K_R\, \mathbf{t} = \mathbf{e}'$$

$$P_R\, X(\infty) = K_R[\, R \mid \mathbf{t} \,] \begin{bmatrix} K_L^{-1}\tilde{\mathbf{x}} \\ 0 \end{bmatrix} = K_R R\, K_L^{-1}\tilde{\mathbf{x}}$$

The first of these is the **epipole** $\mathbf{e}' = K_R\mathbf{t}$, as it must be, since it is the image of the left camera's center.

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Reduction of the construction to the matrix F

The epipolar line in the right view joins the two projected points, and is therefore given by a **cross product**:

$$\ell' = \underbrace{(K_R \mathbf{t})}_{\mathbf{e}'} \times \underbrace{(K_R R\, K_L^{-1} \tilde{\mathbf{x}})}_{\text{second point}} = (K_R \mathbf{t}) \times \big(K_R\,(R K_L^{-1} \tilde{\mathbf{x}})\big).$$

Both terms carry a factor K_R, so we apply the identity $(M\mathbf{a}) \times (M\mathbf{b}) = M^{-\top}(\mathbf{a} \times \mathbf{b})$, valid up to scale, with $M = K_R$:

$$\ell' = K_R^{-\top}\big(\mathbf{t} \times R K_L^{-1} \tilde{\mathbf{x}}\big) = K_R^{-\top}\,[\mathbf{t}]_\times\, R\, K_L^{-1}\,\tilde{\mathbf{x}}.$$

From this we read off the two matrices of interest:

$$\boxed{E = [\mathbf{t}]_\times R} \quad \text{(essential)} \qquad \boxed{F = K_R^{-\top}\,[\mathbf{t}]_\times R\, K_L^{-1} = K_R^{-\top} E\, K_L^{-1}} \quad \text{(fundamental)}$$

$$\ell' = F\,\tilde{\mathbf{x}}, \qquad \text{and a correct match satisfies } \; \tilde{\mathbf{x}}'^{\top} F\, \tilde{\mathbf{x}} = 0.$$

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What the matrix F represents

The matrix F maps a **point** in the left image to a **line** in the right image, namely its epipolar line. This is unlike a homography, which maps a point to a point.

The direction of the mapping matters.

$$\ell' = F\,\tilde{\mathbf{x}} \quad \text{from left to right}$$

$$\ell = F^{\top}\tilde{\mathbf{x}}' \quad \text{from right to left}$$

The matrix F acts from P_L to P_R, and its **transpose** $F^{\top}$ acts from P_R to P_L. The geometry is the same, read in the opposite direction.

The constraint appears symmetric but expresses incidence. The relation $\tilde{\mathbf{x}}'^{\top} F \tilde{\mathbf{x}} = 0$ states that $\mathbf{x}'$ lies on the line $F\tilde{\mathbf{x}}$. It expresses the incidence of a point with a **line** and is not a map from point to point. Given $\mathbf{x}$ alone, F does not determine $\mathbf{x}'$; it determines the **line** along which $\mathbf{x}'$ must be sought.

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Properties of F and E

The epipoles are the null vectors. Every epipolar line $F\tilde{\mathbf{x}}$ passes through $\mathbf{e}'$, so that $\mathbf{e}'^{\top} F\tilde{\mathbf{x}} = 0$ for **all** $\tilde{\mathbf{x}}$. Likewise every left epipolar line $F^{\top}\tilde{\mathbf{x}}'$ passes through $\mathbf{e}$, so that $\tilde{\mathbf{x}}'^{\top} F\,\mathbf{e} = 0$ for **all** $\tilde{\mathbf{x}}'$. A vector orthogonal to every $\tilde{\mathbf{x}}$ must be zero, and hence

$$F^{\top}\mathbf{e}' = \mathbf{0}, \qquad F\,\mathbf{e} = \mathbf{0}.$$

The right null vector $\mathbf{e}$ and the left null vector $\mathbf{e}'$ are the two epipoles.

Rank and degrees of freedom. The matrix F is 3×3 and is defined up to scale, which leaves eight parameters. It has **rank 2**, since a null vector exists, and this imposes one further constraint, leaving **seven degrees of freedom**. In particular $\det F = 0$.

The matrix E has a special structure. Since $E = [\mathbf{t}]_{\times} R$ is the product of a skew-symmetric matrix with a rotation, it has

- **rank 2**, and
- singular values of the form $(\sigma, \sigma, 0)$, two being equal and one zero.

The relation between F and E. We have $F = K_R^{-\top} E\, K_L^{-1}$. Removing the intrinsic parameters, by replacing $\tilde{\mathbf{x}}$ with $K^{-1}\tilde{\mathbf{x}}$ so that the rays are calibrated, reduces F to E. Thus F operates on **pixel** coordinates and E on **calibrated** coordinates.

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References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See **Ch. 9, "Epipolar Geometry and the Fundamental Matrix"**, in particular §9.2 on F, §9.6 on E and §9.6.2 on the $(\sigma, \sigma, 0)$ property.
2. D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, 2nd ed. Pearson, 2012. See Ch. 7, "Stereopsis", on the epipolar constraint and the essential and fundamental matrices.

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